Networks & Distributed Programming

The Application Layer: DNS

Lecture 7



UNIVERSITÀ DEGLI STUDI DI NAPOLI
FEDERICO II



DIETI

UNINA

Application Layer
DNS

- Every exchange we have followed so far started after somebody already knew an address.
- A **URL** carries a **name**, and no TCP connection can be opened towards a name.
- Delivering mail means finding the machine that accepts it for a domain, which is again a name.
- Each time we said: the address is supplied by the DNS (Domain Name System) …

Application Layer
DNS

- There are two ways to identify a host: by **hostname and by IP address**. People prefer the more mnemonic hostname identifier, while network devices prefer fixed-length, hierarchically structured IP addresses.
  - Example: www.unina.it -> 143.225.15.50

- The **Domain Name System** (DNS) is an application-layer protocol that provides distributed name resolution, including the mapping of hostnames to IP addresses.

- It is a **client-server protocol** in which a DNS-client asks to a DNS-server for a specific hostname-to-address translation.

- DNS typically uses **UDP port 53** for queries, but can also use TCP port 53

- Several DNS server implementations exist; BIND (Berkeley Internet Name Domain) is widely used DNS on Linux and Unix-like systems.

The Domain Name System (DNS)

- Mechanism to “convert” names to IP addresses and related other information (mail server, name server, etc.)
- Impossible to manage a centralised repository (table)
  - large number of hosts
  - geographical distance of hosts and delays
- DNS is a distributed, decentralised system

The Domain Name System (DNS)

- Domain Name System
  - A **distributed database**, implemented in a hierarchy of servers, and the application-layer protocol that lets hosts query it
  - It is **client-server**: a client asks for one translation, a server answers.
  - Queries and replies travel in **UDP datagrams**, on **port 53**.

It is the one application nobody uses on purpose: it works for other applications, and only shows up when it fails.

DNS (cont'd)

- Suppose one machine held every name-to-address pair in the world.
  - **Single point of failure**: if it crashes, every application that resolves a name stops with it.
  - **Traffic**: it would answer for every Web request and every mail message sent on the planet.
  - **Distance**: it cannot be close to everybody: from the other side of the world every lookup pays the trip.
  - **Maintenance**: one record per host, and hosts are added and moved continuously.

The DNS is distributed by design: hierarchical, domain-based, and nowhere complete.

DNS (cont'd)

- hierarchical
- domain-based
- implemented by means of a distributed database

Resolver
- The system component used by applications to request name resolution.
- It builds the DNS query and obtains the answer.
- It relies on a configured DNS server to perform the resolution.

The DNS name space

- **The DNS namespace** is a **hierarchical tree** of **domain names.**
- Each node in the tree represents a domain name.
- Naming hierarchy organised by ICANN (Internet Corporation for Assigning Names and Numbers)
- The first level below the root consists of Top-Level Domains (TLDs)
- Each domain partitioned in sub-domains, and so on

Top-Level Domains

Two types

- **generic** (aero, com, edu, . . .)
- **country** (uk, uk, it, tv, . . .)

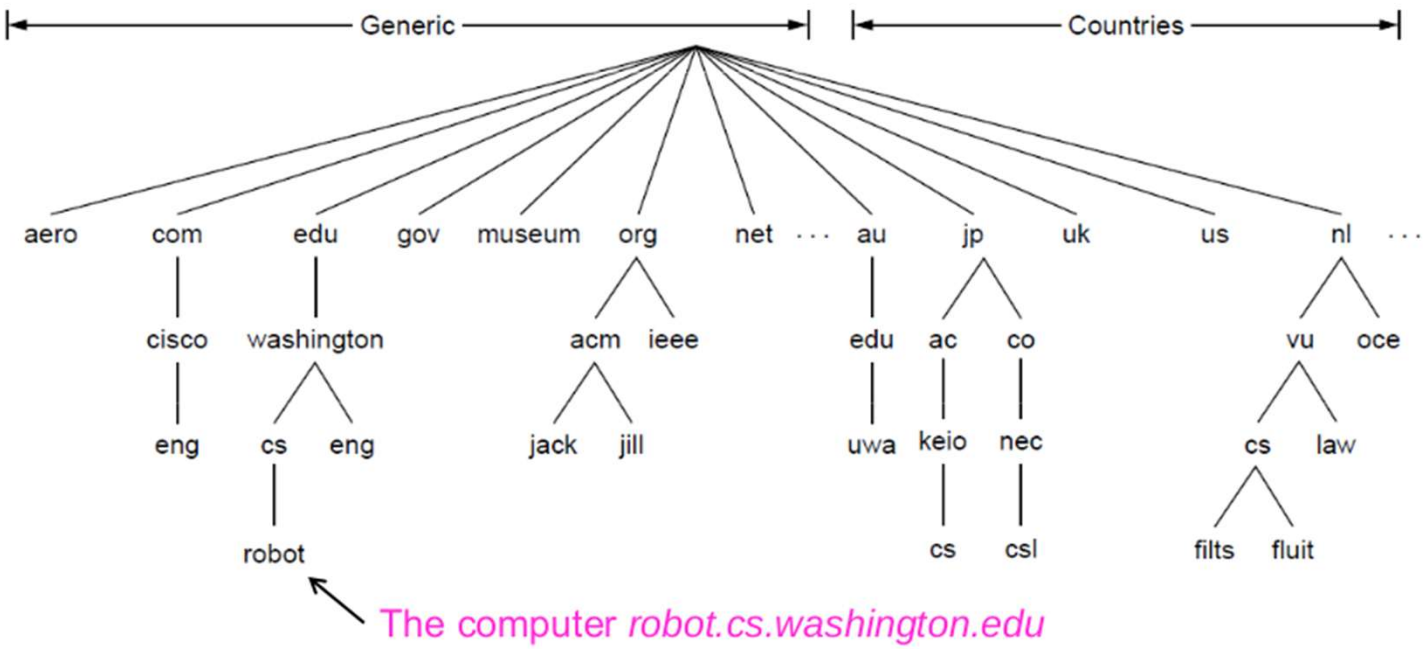
all managed by registrars appointed by ICANN

The DNS name space

- The name space is hierarchical from the root down
- Different parts delegated to different organisations

FQDN = Fully Qualified Domain Name
www.mit.edu.

DN = Domain Name
www.mit.edu



The DNS name space

- **ICANN and the Delegation of Top-Level Domains**
  - From U.S.-backed technical coordination to a global delegation model
  - **1969–1980s | The origins.**
    - Jon Postel and IANA (Internet Assigned Numbers Authority) progressively took responsibility for coordinating key Internet identifiers: IP addresses, protocol parameters and, with the emergence of the DNS, the root and **Top-Level Domains**.
  - **1983–1998 | The DNS takes shape**
    - TLDs such as .com, .org, .net and .edu were established alongside country-code TLDs (it, .uk, .fr.)
    - IANA operated within a U.S. Government-funded and contracted framework.
  - **1998–2016 | ICANN is established**
    - Transition toward a private, nonprofit and internationally participated governance model.
    - ICANN established in 1998 and took responsibility for the IANA functions.
    - Transition completed in 2016, end of the U.S. Government's special oversight role over IANA.

The DNS name space (cont'd)

Generic top-level domains are controlled by ICANN who appoints registrars to run them

Domain	Intended use	Start date	Restricted?
com	Commercial	1985	No
edu	Educational institutions	1985	Yes
gov	Government	1985	Yes
int	International organizations	1988	Yes
mil	Military	1985	Yes
net	Network providers	1985	No
org	Non-profit organizations	1985	No
aero	Air transport	2001	Yes
biz	Businesses	2001	No
coop	Cooperatives	2001	Yes
info	Informational	2002	No
museum	Museums	2002	Yes
name	People	2002	No
pro	Professionals	2002	Yes
cat	Catalan	2005	Yes
jobs	Employment	2005	Yes
mobi	Mobile devices	2005	Yes
tel	Contact details	2005	Yes
travel	Travel industry	2005	Yes
xxx	Sex industry	2010	No

Seven of them are as old as the DNS itself; the rest were opened in waves from 2001 on.

***Restricted** means that a registrar has to check that you are entitled to the name — a museum, an air carrier — before selling it*

The DNS name space

- ICANN /IANACoordinates the DNS root and delegates TLDs

- **Registry operator** — manages a TLD (manages the register),
  - e.g. Verisign → .com, PIR → .org | Registro.it → .it
- **Registrar** — sells domain registrations,
  - e.g. GoDaddy, Namecheap, OVHcloud
- **Registrant** — Domain holder: registers and controls a domain,
  - e.g. Company → company.com

- **Registrant controls all subdomains**
  - *www.company.com* | *shop.company.com* | *api.company.com*

Key principle: authority is delegated down the DNS hierarchy.

The DNS name space (cont'd)

The DNS name space: before the Web

- DNS predates the World Wide Web and HTTP
- Domain names were already used to identify hosts and services on the Internet
- The Web later reused DNS names inside URLs
- A domain name identifies a host or domain, while a URL identifies a specific resource and also specifies how to access it
  - www.example.com→ domain/host name
  - https://www.example.com/course/index.html→ URL

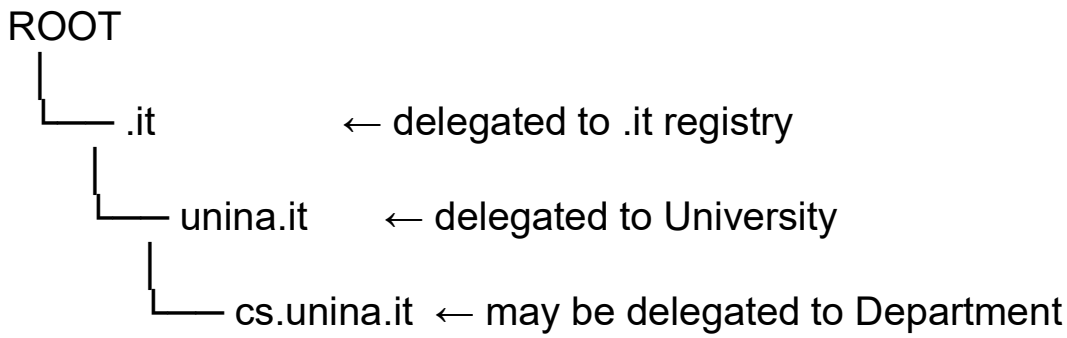
Domain names

- Names are paths upwards from the domain to the root
  - example: eng.mit.edu
- Absolute paths end with a dot (Fully Qualified Domain Name); relative paths are to be interpreted (relative domain name)

- Possibility of registering under more than one top-level domain
  - example: ibm.com, ibm.us
- cyber-squatting: buying domains only in order to sell them later to interested businesses

Domain names (cont'd)

- The administrator of a domain controls the names immediately below it.
- A subdomain can be delegated to another administrator
  - The new administrator can then create and delegate names below that subdomain
- Authority follows the DNS tree: each level controls the namespace delegated to it.



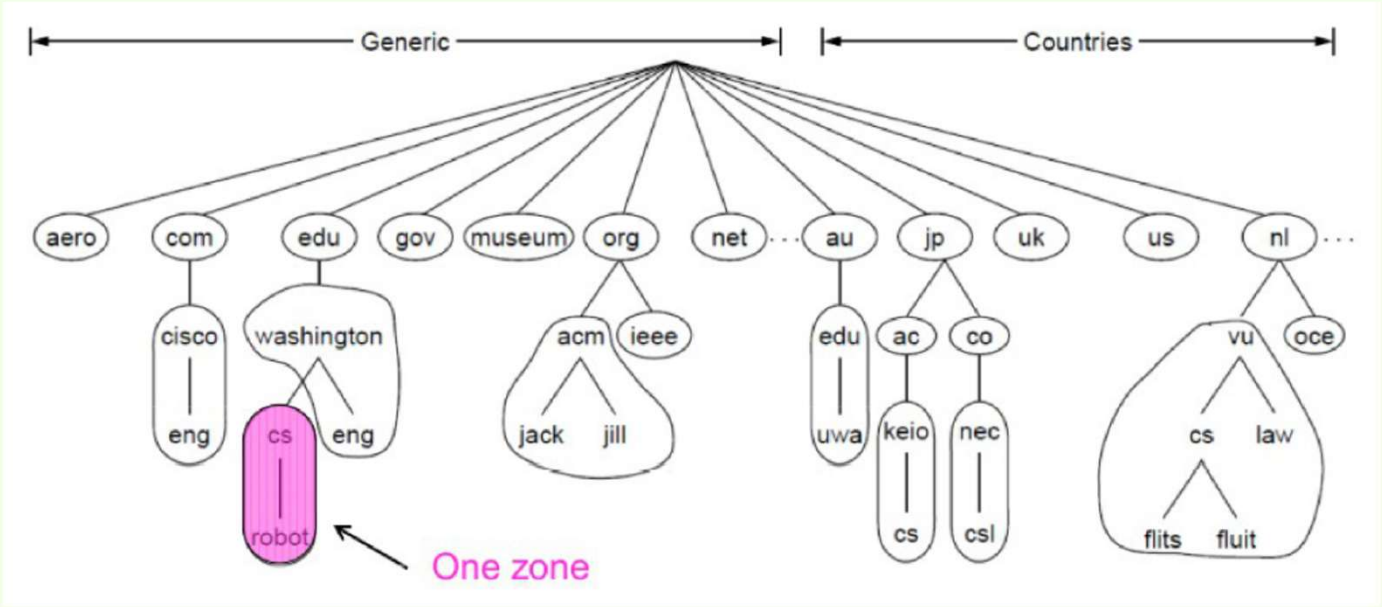
Domain names (cont'd)

- **Zone**
  - A domain is a node in the DNS tree and all its descendants.
  - A **zone** is the portion of that domain **managed by the same DNS authority**.
  - Delegating a subdomain creates a new zone, but the subdomain remains part of the parent domain.
    - Es. If cs.unina.it is delegated, it becomes a separate zone, while remaining part of the unina.it domain.

Domain = naming hierarchy · Zone = administrative authority.

Domain = a node and all its descendants in the DNS namespace.
Zone = a portion of that namespace under the same DNS authority.

Domain names (cont'd)



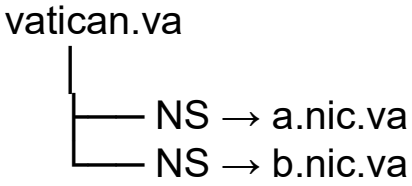
The same tree, cut differently: each outline is a zone, and a zone can be one node or several. No outline ever crosses a delegation.

Domain names (cont'd)

Authoritative name servers

- Each zone is served by one or more authoritative name servers.
- They hold the DNS records for the zone and provide authoritative answers.
- Multiple authoritative servers provide redundancy and availability.
- DNS hosting may be operated directly or provided by a third party.

  - a.nic.va has authority over vatican.va: that is where those records actually live.



Domain names (cont'd)

Authoritative DNS hierarchy

- **Root servers**: Autoritative for ".". They answer with the servers for jp, for edu. E.g., "ask .com servers"
- **TLD servers**: one set per top-level domain, holding the servers of the domains registered under it. E.g., "ask example.com servers"
- **Authoritative name servers**: authoritative for delegated zones below the TLD. Further delegations may create additional levels. E.g., www.example.com → IP

Your host normally queries a recursive resolver, which walks the authoritative hierarchy on its behalf.

Domain resource records

Entry of the distributed database.

Five fields: name, time to live, class, type, value.

- A DNS reply carries one or more records:
  - they are what a server returns, and what a zone is made of.
- The type is what decides how to read the other two:
  - what the name means, and what the value means.
- The database is not stored anywhere as a whole
  - these records, spread over many servers, are the database.

Domain resource records

Fields of a resource record

- domain name
- time to live
- class
- type
- value

Domain resource records

Fields of a resource record

- Name: the domain the record applies to. It is the search key of a query, and several records normally exist for the same name.
- TTL: how long a copy of this record may be kept elsewhere: `86400` (a day) for something stable, `60` for something that moves.
- The maximum is $2^{31} - 1$ seconds, about 68 years; the value is chosen by whoever administers the zone.
- Class: `IN`, for Internet. Other classes exist and you will not meet them.

Resource record fields: Domain name (NAME)

- Tells the domain to which the record applies
- This field is the main search key for DNS queries
- Normally several records exist for each domain, residing at different servers

Resource record fields: Time to live (TTL)

- Indicates the life time of the record
  - long for stable hosts (e.g. 86, 400 for 1 day), short for volatile ones (e.g. 60 for 1 min)
  - maximum value 231 − 1, approximately **68 years**

Resource record fields: Class (CLASS)

- This field is always IN (Internet) for Internet records
- Other codes are rarely used

Resource record fields: Type (cont'd)

Possible values:

- SOA (Start of Authority): information about the server's zone, administrator's contact etc.
- A (Address): 32-bit IPv4 address of some interface for the host
  - some hosts have more than one interface
  - A value for IPv4 (www.example.com. A 93.184.216.34)
  - equivalent AAAA value for IPv6 addresses (www.example.com. AAAA 2001:db8::42)
- MX (Mail eXchange): gives the name of the host which accepts email in the domain (example.com. MX 10 mail.example.com)
- [see next slide]

Resource record fields: Type (TYPE)

- NS (Name Server): identifies the authoritative name servers for a zone; used for delegation.

  ```
  example.com.   NS   ns1.example.net.
  example.com.   NS   ns2.example.net.
  ```

- CNAME (Canonical Name): makes one DNS name an alias of another; resolution continues with the canonical name

  ```
  shop.example.com.   CNAME   stores.example.net.
  ```

- PTR (Pointer): maps an IP address back to a domain name (reverse DNS)
- ...

Resource record fields: Type (TYPE)

- A / AAAA: name → IP
- PTR: IP → name
- CNAME: name → canonical name
- NS: zone → authoritative name server

- SOA: zone → zone authority information

Resource record fields: Type (TYPE)

Type	Meaning	Value
SOA	Start of Authority	Administrative parameters for this zone
A	IPv4 Address	IPv4 address associated with a name
AAAA	IPv6 Address	IPv6 address associated with a name
MX	Mail Exchange	Priority and mail server for the domain
NS	Name Server	Authoritative name server for this zone
CNAME	Canonical Name	Canonical name to which an alias points
PTR	Pointer	Domain name associated with an IP address (reverse DNS)
SRV	Service	Host and port providing a service
TXT	Text	Arbitrary text data

Resource record fields: Value (RDATA)

- The value depends on the record type (see table in previous slide)
- Can contain a domain, a value, or an ASCII string

- MX: domain → priority + mail server
- SOA: zone → zone authority information
  - MNAME (Primary Name Server)
  - RNAME (Administrator's e-mail)
  - Serial number (to synch with secondary authoritative servers)
  - Refresh (how often secondary servers check for updates)
  - Retry (how long a secondary waits before retrying a failed check)
  - Expire (when secondary servers must stop using an outdated copy)
  - Minimum (how long negative answers can be cached)

Domain resource records: example

Name	TTL	class	Type	RDATA
; Authoritative data for cs.vu.nl				
cs.vu.nl.	86400	IN	SOA	star boss (9527,7200,7200,241920,86400)
cs.vu.nl.	86400	IN	MX	1 zephyr
cs.vu.nl.	86400	IN	MX	2 top
cs.vu.nl.	86400	IN	NS	star
star	86400	IN	A	130.37.56.205
zephyr	86400	IN	A	130.37.20.10
top	86400	IN	A	130.37.20.11
www	86400	IN	CNAME	star.cs.vu.nl
ftp	86400	IN	CNAME	zephyr.cs.vu.nl
flits	86400	IN	A	130.37.16.112
flits	86400	IN	A	192.31.231.165
flits	86400	IN	MX	1 flits
flits	86400	IN	MX	2 zephyr
flits	86400	IN	MX	3 top
rowboat		IN	A	130.37.56.201
		IN	MX	1 rowboat
		IN	MX	2 zephyr
little-sister		IN	A	130.37.62.23
laserjet		IN	A	192.31.231.216

← Name server

← IP addresses of computers

← Mail gateways
Priority + mail server

Zone: cs.vu.nl
A: address
NS: Name Server

86.400 sec = 24 h

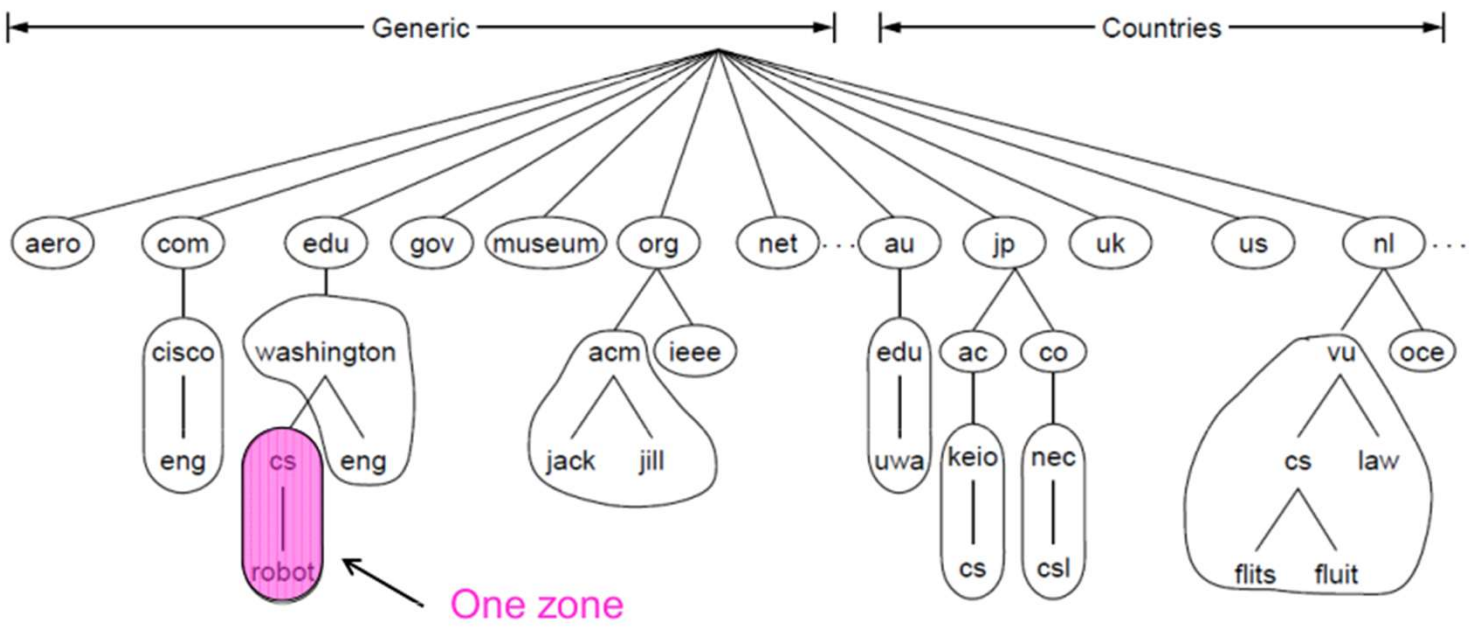
star.cs.vu.nl

Name servers

- A single, centralised DNS server makes no sense
- The DNS name space is divided in non-overlapping zones
- The partitioning is up to the zone administrator
- Each zone contains one or more name servers
  - a primary one and secondary one

Name servers (cont'd)

- Name servers contain data for portions of the name space called zones (circled)



Name resolution

1. Host queries a local name server
2. If the domain falls under the jurisdicion (zone) of the local server, an authoritative record is returned
3. otherwise, the local server handles the query recursively and issues remote queries
   - if the DNS server has the address, it returns it
   - otherwise, it returns the name server for a lower zone
   - the process goes on iteratively, until the address is found

Name resolution (cont'd)

- **Root name servers** are the starting point when a resolver has no useful cached information.
  - There are 13 logical root server identities, named **a.root-servers.net** through **m.root-servers.net**.
- Each identity is implemented by many replicated server instances distributed worldwide.
- Root servers are authoritative for the DNS root zone (.).
- They usually do not return the final record for a host name.
- Instead, they return a referral to the name servers responsible for the relevant **top-level domain**, such as .com, .org, or .edu.

Name resolution (cont'd)

Recursive from the host, iterative from there on

- Recursive query
  - Whoever is asked must produce the final answer, or an error.
    The asker waits and does nothing.
- Iterative query
  - The reply is either the answer or a referral to the next server.
    The asker keeps working.

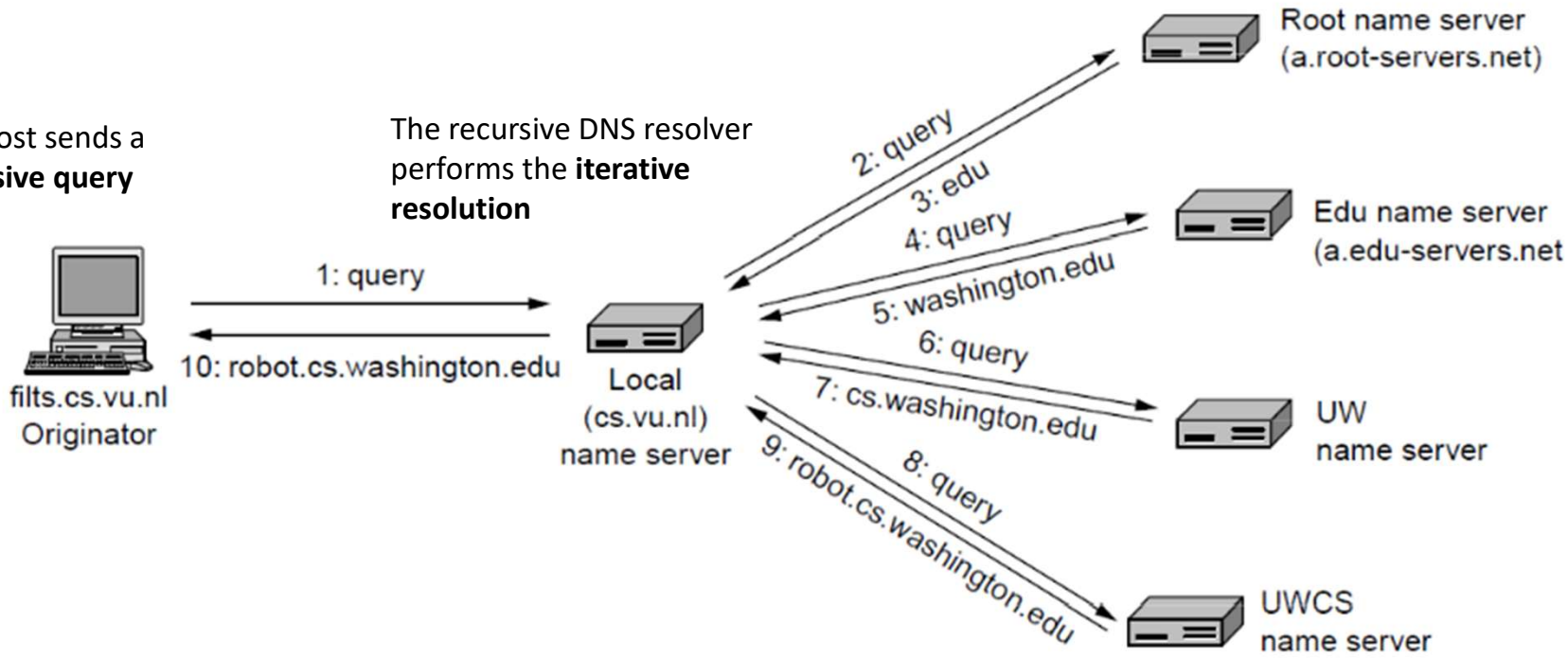
  - The **host** asks once, recursively: one query out, one address back (host).
  - The **local server** does that walk, iteratively, one server at a time (recursive DNS resolver).
  - Root, TLD, and authoritative DNS servers normally answer iteratively: they return an authoritative answer or a referral, rather than performing recursion for the client (Root/TLD/authoritative DNS).

Name servers: example

Root/TLS/authoritative DNS answer **iterative queries**

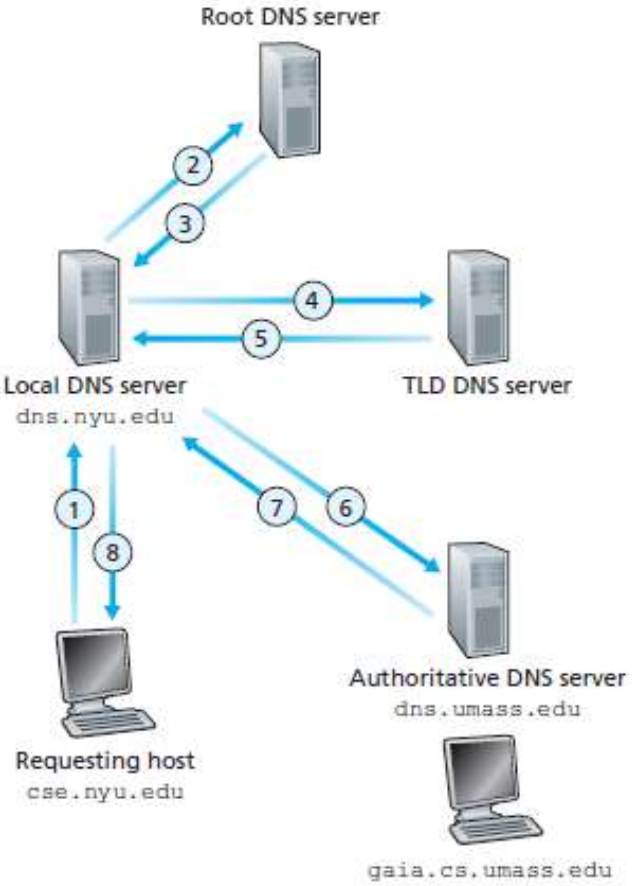
The host sends a **recursive query**

The recursive DNS resolver performs the **iterative resolution**



DNS Resolver

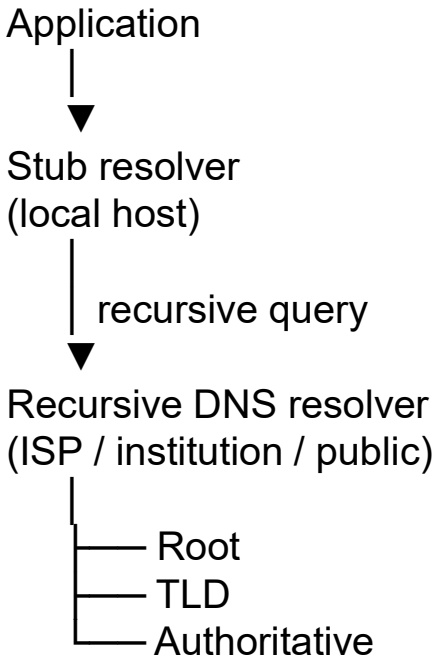
- The **DNS resolver** is a recorsive DNS server.
- Hosts are configured to use one or more recursive DNS resolvers, often provided by the ISP, the local network, or a public DNS service:
  - Google also provides similar servers all around the world (Google Public DNS) on address 8.8.8.8 or 8.8.4.4.
- When a host connects to an ISP, the ISP provides the host with the IP addresses of **one or more of its local DNS servers** (that are typically "close to" the host).
- When a host makes a DNS query, the query is sent to the local DNS server, which **acts as a proxy**, forwarding the query to the DNS server hierarchy.



Stub resolver

A host may not perform DNS resolution itself.

- A stub resolver on the host handles DNS requests from applications.
- It sends a recursive query to a **configured recursive DNS resolver**.
- The recursive resolver performs the actual resolution and returns the final answer.
- On some systems, a local DNS service acts as the **stub/forwarder** (e.g. 127.0.0.53 on many Linux systems).



nslookup examples

- **nslookup** is a command-line tool for querying the DNS
  - Given a domain name, it can retrieve its IP address and other DNS information
    nslookup www.example.com
- We can query a specific DNS server and manually reproduce the iterative querying done by a recursive DNS resolver
- Reverse lookup is also possible (get name from address)

nslookup examples

nslookup sends the query to the local DNS service at 127.0.0.53:53, which forwards it to the configured recursive DNS resolver.

- The returned records are non-authoritative
- They do not come from the DNS server that manages the zone
- The recursive DNS resolver handles the whole resolution via iterative queries
- The host issues a recursive query (either the complete answer or an error is returned)

To check DNS config:
resolvectl status or cat /etc/resolv.conf

```
andrea@turgia:~$ nslookup www.vatican.va
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
www.vatican.va  canonical name = wcm-disp.spc.va.
Name:   wcm-disp.spc.va
Address: 185.152.70.33
```

DNS Caching

- Every reply a resolver receives can be cached and used to answer subsequent queries.

- **A recursive resolver's answer is non-authoritative**
  - the resolver is not authoritative for the queried zone, whether the answer comes from its cache or has just been resolved.
- The **addresses of TLD servers are cached**
  - a resolution usually starts below the root.
- **Not only addresses (A/AAAA) are cached**
  - but also NS records and other DNS information. This can avoid repeating the whole resolution path.
- **Cached records are kept only for their TTL** (Time To Live)
  - after which they must be refreshed.

Caching is what makes the whole thing work at this scale — and the TTL is what keeps a cached copy from lying for ever.

DNS Caching

- Suppose a resolver needs to resolve www.cs.mit.edu and the final A/AAAA record is not in its cache.
- The resolver uses the most specific useful information already cached:

Cached information		Resolution starts from
A/AAAA for www.cs.mit.edu	→	Answer immediately
NS for cs.mit.edu	→	cs.mit.edu authoritative server
NS for mit.edu	→	mit.edu authoritative server
NS for .edu	→	.edu TLD server
No useful cached delegation	→	Root server

A resolver does not necessarily start from the root: it starts from the closest known point in the DNS hierarchy.

DNS Caching

- Suppose a resolver needs to resolve www.cs.mit.edu and the final A/AAAA record is not in its cache.
- The resolver uses the most specific useful information already cached:

Information already cached	**Resolution starts from**	**New information that may be cached**
NS for cs.mit.edu	cs.mit.edu authoritative server	A/AAAA for www.cs.mit.edu
NS for mit.edu	mit.edu authoritative server	NS for cs.mit.edu + final A/AAAA
NS for .edu	.edu TLD server	NS for mit.edu, NS for cs.mit.edu + final A/AAAA
No useful delegation	Root server	NS for .edu, mit.edu, cs.mit.edu + final A/AAAA

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

- It shows DNS responses in a format close to the actual DNS message structure.
- It can query a specific Resource Record type (A, AAAA, NS, MX, SOA, ...).
- It can query a specific DNS server.
- With +trace, it can show the delegation path through the DNS hierarchy.

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

- dig vatican.va

```
; <<>> DiG 9.18.39-0ubuntu0.24.04.7-Ubuntu <<>> vatican.va
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 34557
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1220
;; QUESTION SECTION:
;vatican.va.                    IN      A

;; ANSWER SECTION:
vatican.va.             86400   IN      A       185.152.70.106

;; Query time: 52 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Sun Oct 04 17:20:18 CEST 2026
;; MSG SIZE  rcvd: 55
```

Header - recursion was requested (rd) and available (ra). No aa flag = response is non-authoritative.
Question - requested record
Answer - returned resource records, including their TTL
Final - query time, the DNS server contacted, protocol, and message size

What a DNS message carries

Queries and replies have the same format: a 12-byte header, then four sections.

- **Header:** an identifier copied into the reply, so the client can match answers to questions, plus a set of flags.
- **The flags that matter:** authoritative answer, recursion desired, recursion available.
- **Question:** the name being asked about, and the type asked for.
- **Answer:** the records that answer it.
- **Authority:** servers that would know.
- **Additional:** helpful records, typically the addresses of the servers just named.

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

- Querying a specific record type:
  - +noall +answer keeps only the answer section
  - A, AAAA, NS, MX, SOA

dig +noall +answer www.mit.edu A

```
www.mit.edu.                1800    IN      CNAME   www.mit.edu.edgekey.net.
www.mit.edu.edgekey.net.    60      IN      CNAME   e9566.dscb.akamaiedge.net.
e9566.dscb.akamaiedge.net.  20      IN      A       23.50.153.111
```

Two CNAMEs before an address: the published name is an alias for another name, and the machine that finally answers belongs to somebody else entirely. And look at the TTLs: the alias is kept for minutes, the final address for seconds.

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

- Querying a specific record type:
  - +noall +answer keeps only the answer section
  - A, AAAA, NS, MX, SOA

dig +noall +answer mit.edu MX

```
;; communications error to 10.255.255.254#53: timed out
;; communications error to 10.255.255.254#53: timed out
mit.edu.                65      IN      MX      100 mit-edu.mail.protection.outlook.com.
```

mit.edu. 65 IN MX 100 mit-edu.mail.protection.outlook.com.

↑ TTL ↑ TYPE ↑ preference ↑ mail server

MX RDATA contains two fields: a preference value and the mail server name. Lower preference values have higher priority.

dig — Domain Information Groper

dig is a command-line tool for querying DNS servers and inspecting DNS responses.

- Querying a specific record type:
  - +noall +answer keeps only the answer section
  - A, AAAA, NS, MX, SOA

dig +noall +answer mit.edu NS

```
mit.edu.                1800    IN      NS      asia2.akam.net.
mit.edu.                1800    IN      NS      use5.akam.net.
mit.edu.                1800    IN      NS      usw2.akam.net.
mit.edu.                1800    IN      NS      use2.akam.net.
mit.edu.                1800    IN      NS      ns1-37.akam.net.
mit.edu.                1800    IN      NS      asia1.akam.net.
mit.edu.                1800    IN      NS      eur5.akam.net.
mit.edu.                1800    IN      NS      ns1-173.akam.net.
```

NAME	TTL	CLASS	TYPE	RDATA
mit.edu.	1800	IN	NS	asia2.akam.net

authoritative name server

An NS query may return multiple records: together they form an NS RRset.
Each record identifies an authoritative name server for the zone.